

How to write an R Markdown file

A short guide, using lecture03_tidyverse.Rmd as the example

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1 What an R Markdown file is

An R Markdown file (`.Rmd`) is a text file that mixes three things:

1. a **header** at the top, between two lines of `---`, that says what the document is called and what it should become (PDF or web page);
2. **text**, written in Markdown, a plain way of marking headings, bold and lists;
3. **code chunks**, blocks of R code that run when the document is *knitted*, with their output placed right after them.

Pressing **Knit** in RStudio turns the file into a finished document: your explanation, your code and its results, together. That is why assignments should use it: the reader sees exactly what you ran and what came out.

This guide is itself an R Markdown file. Open `rmarkdown_guide.Rmd` next to the PDF and compare them; every part described below is visible in the source.

2 Create one

In RStudio: **File** → **New File** → **R Markdown...** Give it a title and your name, choose **PDF** or **HTML**, click **OK**. RStudio opens a file with example content. Delete everything below the header and write your own.

Save it with **Ctrl + S**, in a folder of its own, with any data in a `data/` subfolder next to it. When knitting, R uses the folder of the `.Rmd` file as its working directory, so `read_csv("data/students.csv")` finds the file with no `setwd()`.

3 The three parts, using the lecture file

3.1 The header

The top of `lecture03_tidyverse.Rmd` looks like this:

```
---
title: "Lecture 3: Introduction to the Tidyverse"
author: "Touseef Hameed"
date: "16 September 2026"
output: pdf_document
---
```

Change the title, author and date. `output: pdf_document` makes a PDF; `output: html_document` makes a web page. Keep the indentation exactly as shown; the header is sensitive to spaces.

3.2 Text

Ordinary sentences are just typed. A few marks give structure:

```
# A main heading
## A smaller heading
```

Plain paragraph. **Bold**, *italic*, and `code in the text`.

- a bullet point
- another one

1. a numbered step
2. the next step

A link: [<https://r4ds.hadley.nz>](https://r4ds.hadley.nz)

In an assignment, use one # heading per problem, then the code, then your explanation.

3.3 Code chunks

A chunk starts with ````{r}` on its own line and ends with `````:

```
``` r
library(tidyverse) # load the packages
```

```
#> Warning: package 'tidyverse' was built under R version 4.4.3
```

```
#> Warning: package 'ggplot2' was built under R version 4.4.3
```

```
#> Warning: package 'tibble' was built under R version 4.4.3
```

```
#> Warning: package 'tidyr' was built under R version 4.4.3
```

```
#> Warning: package 'readr' was built under R version 4.4.2
```

```
#> Warning: package 'dplyr' was built under R version 4.4.3
```

```
#> Warning: package 'lubridate' was built under R version 4.4.2
```

```
#> -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
#> v dplyr 1.2.0 v readr 2.1.5
#> v forcats 1.0.0 v stringr 1.5.1
#> v ggplot2 4.0.0 v tibble 3.3.1
#> v lubridate 1.9.4 v tidyr 1.3.1
#> v purrr 1.0.2
```

```

#> -- Conflicts ----- tidyverse_conflicts() --
#> x dplyr::filter() masks stats::filter()
#> x dplyr::lag() masks stats::lag()
#> i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
...

``` r
students <- read_csv("data/students.csv") # read the data
...

...
#> Rows: 6 Columns: 5
#> -- Column specification -----
#> Delimiter: ","
#> chr (4): Full Name, favourite.food, mealPlan, AGE
#> dbl (1): Student ID
#>
#> i Use `spec()` to retrieve the full column specification for this data.
#> i Specify the column types or set `show_col_types = FALSE` to quiet this message.
...

``` r
head(students) # first six rows
...

...
#> # A tibble: 6 x 5
#> `Student ID` `Full Name` favourite.food mealPlan AGE
#> <dbl> <chr> <chr> <chr> <chr>
#> 1 1 Sunil Huffmann Strawberry yoghurt Lunch only 4
#> 2 2 Barclay Lynn French fries Lunch only 5
#> 3 3 Jayendra Lyne N/A Breakfast and lunch 7
#> 4 4 Leon Rossini Anchovies Lunch only <NA>
#> 5 5 Chidiegwu Dunkel Pizza Breakfast and lunch five
#> 6 6 Güvenç Attila Ice cream Lunch only 6
...

```

Insert a new chunk with **Ctrl + Alt + I**. Run one chunk without knitting by clicking the green **play** button at its top right. Its output appears under it in the editor.

When the document is knitted, that chunk produces this:

```

library(tidyverse)
students <- read_csv("data/students.csv", show_col_types = FALSE)
head(students)

```

```

#> # A tibble: 6 x 5
#> `Student ID` `Full Name` favourite.food mealPlan AGE
#> <dbl> <chr> <chr> <chr> <chr>

```

```
#> 1 1 Sunil Huffmann Strawberry yoghurt Lunch only 4
#> 2 2 Barclay Lynn French fries Lunch only 5
#> 3 3 Jayendra Lyne N/A Breakfast and lunch 7
#> 4 4 Leon Rossini Anchovies Lunch only <NA>
#> 5 5 Chidiegwu Dunkel Pizza Breakfast and lunch five
#> 6 6 Güvenç Attila Ice cream Lunch only 6
```

## 4 Chunk options you will need

Options go inside the braces after `r`, separated by commas.

option	effect	use it for
<code>eval=FALSE</code>	show the code, do not run it	<code>install.packages("tidyverse")</code> , which must not run at every knit
<code>echo=FALSE</code>	run the code, hide it, show only the output	a plot where the code is not the point
<code>include=FALSE</code>	run the code, show nothing	a setup chunk
<code>message=FALSE,</code> <code>warning=FALSE</code>	hide messages and warnings	the long message that <code>library(tidyverse)</code> prints

Example, as written in the lecture file:

```
``` r
install.packages("tidyverse") # run once on each computer you use
```
```

## 5 Knit

Click **Knit** (the ball-of-wool icon above the editor), or press **Ctrl + Shift + K**. The document is built in a **fresh R session**: nothing from your console is available. Every package must be loaded and every object created *inside* the document, in order.

The result is saved next to the `.Rmd` file, with the same name and `.pdf` or `.html`.

PDF needs a LaTeX installation. If knitting to PDF fails, either run once in the console

```
install.packages("tinytex")
tinytex::install_tinytex()
```

or change the header to `output: html_document`. HTML is accepted for the assignment.

## 6 The errors everyone meets

| message                                                                                    | cause                                                                                                                                                | fix                                                                                     |
|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| object 'students' not found<br>could not find function "read_csv"                          | the chunk that creates it is missing, or comes later in the file<br><code>library(tidyverse)</code> is not in the document, or comes after the chunk | create every object before you use it, inside the document<br>put it in the first chunk |
| cannot open file 'data/students.csv'<br>the knit stops at<br><code>install.packages</code> | the file is not in a <b>data</b> folder next to the <code>.Rmd</code><br>the chunk is being run                                                      | move it there, or use the web address<br>add <code>eval=FALSE</code> to that chunk      |
| Error in parse: unexpected symbol                                                          | a typing error in the chunk, often a missing comma or quote                                                                                          | run the chunk alone with the play button to find the line                               |

The knitting log, in the **R Markdown** tab at the bottom of RStudio, names the chunk where the problem is.

## 7 For the assignment

- One heading per problem: `# Problem 1`, `# Problem 2`, and so on.
- Under each: the chunk with the code and, for the written problems, your text.
- A comment on every command.
- Knit it, check that the output under each chunk is what you meant, and upload the `.Rmd` together with the knitted `.pdf` or `.html`.

If knitting does not work on your computer, screenshots of your RStudio window showing the code and its output are accepted instead.